

Boolean conditions

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

From a paper algorithm to Python

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace separate checks for Boolean conditions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Boolean conditions. Proposed explanation: "At most excludes the boundary". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Combine requirements without replacing AND

Predict all four booleans. Eligibility requires science AND at most 100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90

genre = "science"

available = False

print(genre == "science" and pages <= 100)

print((genre == "science" and pages <= 100) and available)
```

```
print(not available)

print(genre == "science" or pages <= 100)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Translate the requirement

Write a condition for an activity taking at most 12 minutes and being quiet. Explain why exactly 12 minutes is included.

My reasoning and evidence

6. W2 Trace separate checks

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time comparison and final suitability values before running the program.

My reasoning and evidence

7. W3 Distinguish alternatives

Complete all four Boolean combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

My reasoning and evidence

8. W4 Write a new rule independently

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False). Choose a case that detects replacing and with or between the range check and available.

My reasoning and evidence

9. W5 Separate a check from a guarantee

Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled boolean conditions evidence sheet to Python readiness portfolio. Show how you can translate stated requirements into comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Boolean conditions

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Python 3: `age=15; allowed=(age>=12) and (age<=15); print(allowed)`. No type validation is implemented here.

Boolean conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check lower boundary

$15 \geq 12$ is True.



2. Check upper boundary

$15 \leq 15$ is also True.



3. Combine with and

Both comparisons are True, so allowed is True.

CHANGE ONE THING

Change age to 16.

Explain the changed result using the matching lesson.

Give both comparison results for ages 11, 15 and 16. Which comparison rejects each outside value, and why does and matter?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Both comparisons are True, so allowed is True.

1. Check lower boundary
2. Check upper boundary
3. Combine with and

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Combine with and Both comparisons are True, so allowed is True. A Boolean condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Practice 2: explain the deciding evidence

Changed-case practice: Change age to 16. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The result becomes False.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The result becomes False. The lower condition remains True, but the upper condition fails.

Repair a missing building block

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions. Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
